

KIERSTEN WINTER

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OBJECTIVE

Aspiring Data Analyst with a strong foundation in applied statistics. Experienced in developing computational workflows for large-scale data comparison and performing statistical analysis to identify key insights. Skilled in data cleaning, visualization, and statistical modeling using the tools R and Python.

EDUCATION

ROCHESTER INSTITUTE OF TECHNOLOGY

Bachelor of Science Degree in Applied Statistics and Data Analytics (May 2024)

- Cumulative GPA: 3.560/4.0; Cum Laude
- Minor in Cellular and Molecular Biology
- College of Science Dean's List Fall 2021, Fall 2022, Fall 2023, Spring 2024
- Relevant Courses: Cell Biology; Data Mining; Developmental Biology; Introduction to Data Visualization; Mathematical Statistics I & II; Molecular Biology; Nonparametric Statistics; Statistical Analysis for Bioinformatics; Statistical Software - R; Virology

Master of Science Degree in Applied Statistics (May 2025)

- Cumulative GPA: 4.0/4.0
- Relevant Courses: Applied Linear Models ANOVA & Regression; Causal Inference; Design & Analysis of Clinical Trials; Foundations of Statistics; Nonparametric Statistics & Bootstrapping; Principles of Statistical Machine Learning; Statistical Quality Control; Survey Design & Analysis

WORK EXPERIENCE

ROCHESTER INSTITUTE OF TECHNOLOGY

Graduate Writing Consultant (Aug 2024 - May 2025)

- Provided individualized support to students at any stage of the writing process across various disciplines.

Campus Tour Guide Trainer (Jan 2024 - May 2025)

- Trained new campus tour guides to provide engaging and informative tours to prospective students.
- Ensured effective communication, consistency, and excellence in new tour guides.

Research Assistant (Jan 2024 - Sept 2024)

- Collaborated on research using statistical modeling and analysis to transform protein interactions into resonance, enhancing understanding of molecular dynamics.
- Developed modeling technique that identifies significant vibrational frequencies in molecular dynamics.

Research Data Analyst (Jan 2023 – Jan 2025)

- Developed computational workflow using R that allows for the large-scale comparison of mass spectrometry-based proteomics data.
- Performed statistical analysis and data visualization to identify differentially abundant proteins.

Campus Tour Guide (Jan 2021 – May 2025)

- Assisted the RIT Undergraduate Admissions Office by providing campus tours to prospective students.

SKILLS

- **Technical Skills:** Data cleaning, visualization, interpretation; statistical modeling & analysis; technical writing; beginner proficiency in Korean
- **Software:** LaTeX, JMP, Minitab, R, Python, Excel, SAS

PROJECTS

A Machine Learning Approach to Predict Antibiotic Resistance in *Salmonella* Newport (Jan 2025 - May 2025)

- Developed and evaluated machine learning models to predict antibiotic resistance in *Salmonella* Newport.
- Implemented various techniques to address class imbalance and improve detection of resistance cases.

Forecasting Epidemiology: Time Series Modeling of the West Nile Virus (Jan 2024 - May 2024)

- Conducted comprehensive research on arboviral time series modeling and forecasting.
- Analyzed spatiotemporal variation and influence of environmental factors on case outbreaks.

PUBLICATIONS

- Chatterjee, P., Consoli, C. E., Schiller, H., Winter, K. K., McCallum, M. E., Schulze, S., & Pohlschroder, M. (2025). Quorum sensing mediates morphology and motility transitions in the model archaeon *Haloferax volcanii*. *bioRxiv*. <https://doi.org/10.1101/2025.01.14.633064>.

HONORS

- Received 1st place awards in application at the 2024 *and* 2025 Upstate New York Statistics (UPSTAT) conference.
- Presented at the 2024 Women in Statistics and Data Science (WSDS) conference.
- Nominated for RIT's 2025 College of Science (COS) Graduate Student Delegate.